**VLAN**

show vlan brief, show vlan summary, show interfaces fa0/18 switchport

Normal Range ID: 1 – 1005 (1002-1005 reserved)

Extended Range ID: 1006 – 4096

**Založení VLAN:**

S3(config)# vlan 20

S3(config-vlan)# name student

**S3(config-vlan)# vlan 150**

**S3(config-vlan)# name VOICE**

**Přiřazení portu do VLAN:**

S3(config)# interface fa0/18

S3(config-if)# switchport mode access

S3(config-if)# switchport access vlan 20

**S3(config-if)# mls qos trust cos**

**S3(config-if)# switchport voice vlan 150**

**Vyhození portu z VLAN:**

S1(config)# interface fa0/18

S1(config-if)# no switchport access vlan

S1(config-if)# end

**Vymazání SWICH:**

erase startup-config

delete vlan.dat

**Nastavení portu jako TRUNK:**

S1(config)# interface fastEthernet 0/1

S1(config-if)# switchport mode trunk

S1(config-if)# switchport trunk native vlan 99

S1(config-if)# switchport trunk allowed vlan 10,20,30,99

S1(config-if)# end

**Vyhození portu z TRUNK a NATIVE VLAN:**

S1(config)# interface fa0/1

S1(config-if)# no switchport trunk allowed vlan

S1(config-if)# no switchport trunk native vlan

S1(config-if)# end

**DTP:**

S1(config-if)# switchport mode trunk

S1(config-if)# switchport nonegotiate

---

**Inter-VLAN Routing: Router-on-a-Stick**

1/ Create and name the VLANs

S1(config)# vlan 99

S1(config-vlan)# name Management

S1(config-vlan)# exit

2/ Create the management interface

S1(config)# interface vlan 99

S1(config-if)# ip add 192.168.99.2 255.255.255.0

S1(config-if)# no shut

S1(config-if)# exit

S1(config)# ip default-gateway 192.168.99.1

3/ Configure access ports

S1(config)# interface fa0/6

S1(config-if)# switchport mode access

S1(config-if)# switchport access vlan 10

S1(config-if)# no shut

4/ Configure trunking ports

S1(config)# interface fa0/1

S1(config-if)# switchport mode trunk

S1(config-if)# no shut

S1(config-if)# exit

R1(config)# interface G0/0/1.10

R1(config-subif)# description Default Gateway for VLAN 10

R1(config-subif)# encapsulation dot1Q 10

R1(config-subif)# ip add 192.168.10.1 255.255.255.0

R1(config-subif)# exit

R1(config)# interface G0/0/1.99

R1(config-subif)# description Default Gateway for VLAN 99

R1(config-subif)# encapsulation dot1Q 99

R1(config-subif)# ip add 192.168.99.1 255.255.255.0

R1(config-subif)# exit

R1(config)# interface G0/0/1

R1(config-if)# description Trunk link to S1

R1(config-if)# no shut

R1(config-if)# end

show interfaces trunk

show interfaces

show ip interface brief

---

**Inter-VLAN Routing: Layer 3 Switch**

1/ Create the VLANs

MLS(config)# vlan 10

MLS(config-vlan)# name LAN10

2/ Create the SVI VLAN interfaces

MLS(config)# interface vlan 10

MLS(config-if)# description Default Gateway SVI for 192.168.10.0/24

MLS(config-if)# ip add 192.168.10.1 255.255.255.0

MLS(config-if)# no shut

MLS(config-if)# exit

3/ Configure access ports

MLS(config)# interface GigabitEthernet1/0/6

MLS(config-if)# description Access port to PC1

MLS(config-if)# switchport mode access

MLS(config-if)# switchport access vlan 10

MLS(config-if)# exit

4/ Enable IP routing

MLS(config)# ip routing

5/ Configure the routed port

MLS(config)# interface GigabitEthernet0/0/1

MLS(config-if)# description routed Port Link to R1

MLS(config-if)# no switchport

MLS(config-if)# ip address 10.10.10.2 255.255.255.0

MLS(config-if)# no shut

---

**EtherChannel**

**PAgP – on, desirable, auto**

**LACP - on, active, passive**

S1(config)# **interface range FastEthernet 0/1 - 2**

S1(config-if-range)# **channel-group 1 mode active**

Creating a port-channel interface Port-channel 1

S1(config-if-range)# **exit**

S1(config-if)# **interface port-channel 1**

S1(config-if)# **switchport mode trunk**

S1(config-if)# **switchport trunk allowed vlan 1,2,20**

**Verify**

S1# show interfaces port-channel 1

S1# show etherchannel summary

S1# show etherchannel port-channel

S1# show interfaces f0/1 etherchannel

**Rozbití EtherChannel**

S1(config)# **no interface port-channel 1**

---

**DHCPv4 server on R1**

R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9

R1(config)# ip dhcp pool LAN-POOL-1

R1(dhcp-config)# network 192.168.10.0 255.255.255.0

R1(dhcp-config)# default-router 192.168.10.1

R1(dhcp-config)# dns-server 192.168.11.5

R1(dhcp-config)# domain-name example.com

R1(dhcp-config)# end

**Verify**

R1# show run | section dhcp

R1# show ip dhcp binding

R1# show ip dhcp server statistics

C:> ipconfig /all

**Disable/Enable DHCP**

R1(config)# no service dhcp

R1(config)# service dhcp

**DHCPv4 Relay**

R1(config)# interface g0/0/0

R1(config-if)# ip helper-address 192.168.11.6

R1(config-if)# end

**Verify**

C:> ipconfig /release

C:> ipconfig /renew

C:> ipconfig /all

R1# show ip interface g0/0/0

**DHCP client on R2**

R2(config)# interface G0/0/1

R2(config-if)# ip address dhcp

R2(config-if)# no shutdown

---

**DHCPv6 Statelless Server on R1**

R1(config)# ipv6 unicast-routing

R1(config)# ipv6 dhcp pool IPV6-STATELESS

R1(config-dhcpv6)# dns-server 2001:db8:acad:1::254

R1(config-dhcpv6)# domain-name example.com

R1(config-dhcpv6)# exit

R1(config)# interface GigabitEthernet0/0/1

R1(config-if)# description Link to LAN

R1(config-if)# ipv6 address fe80::1 link-local

R1(config-if)# ipv6 address 2001:db8:acad:1::1/64

R1(config-if)# ipv6 nd other-config-flag

R1(config-if)# ipv6 dhcp server IPV6-STATELESS

R1(config-if)# no shut

R1(config-if)# end

**Configure a Stateless DHCPv6 Client**

R3(config)# ipv6 unicast-routing

R3(config)# interface g0/0/1

R3(config-if)# ipv6 enable

R3(config-if)# ipv6 address autoconfig

R3(config-if)# end

R3# show ipv6 interface brief

R3# show ipv6 dhcp interface g0/0/1

**Configure a Stateful DHCPv6 Server**

R1(config)# ipv6 unicast-routing

R1(config)# ipv6 dhcp pool IPV6-STATEFUL

R1(config-dhcpv6)# address prefix 2001:db8:acad:1::/64

R1(config-dhcpv6)# dns-server 2001:4860:4860::8888

R1(config-dhcpv6)# domain-name example.com

R1(config-dhcpv6)#

R1(config)# interface GigabitEthernet0/0/1

R1(config-if)# description Link to LAN

R1(config-if)# ipv6 address fe80::1 link-local

R1(config-if)# ipv6 address 2001:db8:acad:1::1/64

R1(config-if)# ipv6 nd managed-config-flag

R1(config-if)# ipv6 nd prefix default no-autoconfig

R1(config-if)# ipv6 dhcp server IPV6-STATEFUL

R1(config-if)# no shut

R1(config-if)# end

**Configure a Stateful DHCPv6 Client**

R3(config)# ipv6 unicast-routing

R3(config)# interface g0/0/1

R3(config-if)# ipv6 enable

R3(config-if)# ipv6 address dhcp

R3(config-if)# end

R3# show ipv6 interface brief

R3# show ipv6 dhcp interface g0/0/1

**DHCPv6 Relay Agent**

R1(config)# interface gigabitethernet 0/0/1

R1(config-if)# ipv6 dhcp relay destination 2001:db8:acad:1::2 G0/0/0

R1(config-if)# exit

---

**Port Security**

**Secure Unused Ports**

S1(config)# interface range fa0/8 - 24

S1(config-if-range)# shutdown

**Enable Port Security**

S1(config)# interface f0/1

S1(config-if)# switchport port-security

Command rejected: FastEthernet0/1 is a dynamic port.

S1(config-if)# switchport mode access

S1(config-if)# switchport port-security

S1(config-if)# end

**Limit and Learn MAC Addresses**

S1(config)# interface fa0/1

S1(config-if)# switchport mode access

S1(config-if)# switchport port-security

S1(config-if)# switchport port-security maximum 4

S1(config-if)# switchport port-security mac-address aaaa.bbbb.1234

S1(config-if)# switchport port-security mac-address sticky

S1(config-if)# end

**Port Security Aging**

S1(config)# interface fa0/1

S1(config-if)# switchport port-security aging time 10

S1(config-if)# switchport port-security aging type **inactivity**

S1(config-if)# end

**Pozn.**

**Absolute** – The secure addresses on the port are deleted after the specified aging time.

**Inactivity** – The secure addresses on the port are deleted only if they are inactive for the specified aging time.

**Port Security Violation Modes**

S1(config)# interface f0/1

S1(config-if)# switchport port-security violation **restrict**

Security Violation Mode Comparison

| Violation Mode | Discards Offending Traffic | Sends Syslog Message | Increase Violation Counter | Shuts Down Port |
|---|---|---|---|---|
| Protect | Yes | No | No | No |
| Restrict | Yes | Yes | Yes | No |
| Shutdown | Yes | Yes | Yes | Yes |

**Ports in error-disabled State**

S1(config)# **interface fa0/18**

S1(config-if)# **shutdown**

S1(config-if)# **no shutdown**

**Verify Port Security**

S1# show port-security

S1# show port-security interface f0/1

S1# show port-security address

**Steps to Mitigate VLAN Hopping Attacks**

Step 1: Disable DTP (auto trunking) negotiations on non-trunking ports by using the **switchport mode access** interface configuration command.

Step 2: Disable unused ports and put them in an unused VLAN.

Step 3: Manually enable the trunk link on a trunking port by using the **switchport mode trunk** command.

Step 4: Disable DTP (auto trunking) negotiations on trunking ports by using the **switchport nonegotiate** command.

Step 5: Set the native VLAN to a VLAN other than VLAN 1 by using the **switchport trunk native vlan** vlan_number command.

For example, assume the following:

- FastEthernet ports 0/1 through fa0/16 are active access ports
- FastEthernet ports 0/17 through 0/24 are not currently in use
- FastEthernet ports 0/21 through 0/24 are trunk ports.

S1(config)# interface range fa0/1 - 16

S1(config-if-range)# switchport mode access

S1(config-if-range)# exit

S1(config)# interface range fa0/17 - 20

S1(config-if-range)# switchport mode access

S1(config-if-range)# switchport access vlan 1000

S1(config-if-range)# exit

S1(config)# interface range fa0/21 - 24

S1(config-if-range)# switchport mode trunk

S1(config-if-range)# switchport nonegotiate

S1(config-if-range)# switchport trunk native vlan 999

S1(config-if-range)# end

**Mitigate DHCP Attacks**

S1(config)# ip dhcp snooping

S1(config)# interface f0/1

S1(config-if)# ip dhcp snooping trust

S1(config-if)# exit

S1(config)# interface range f0/5 - 24

S1(config-if-range)# ip dhcp snooping limit rate 6

S1(config-if)# exit

S1(config)# ip dhcp snooping vlan 5,10,50-52

S1(config)# end

S1# **show ip dhcp snooping**

**Mitigate ARP Attacks**

S1(config)# ip dhcp snooping

S1(config)# ip dhcp snooping vlan 10

S1(config)# ip arp inspection vlan 10

S1(config)# interface fa0/24

S1(config-if)# ip dhcp snooping trust

S1(config-if)# ip arp inspection trust

S1(config)# ip arp inspection validate src-mac

S1(config)# ip arp inspection validate dst-mac

S1(config)# ip arp inspection validate ip

S1(config)# do show run | include validate

ip arp inspection validate ip

S1(config)# ip arp inspection validate src-mac dst-mac ip

S1(config)# do show run | include validate

ip arp inspection validate src-mac dst-mac ip

**Mitigate STP Attacks**

**Configure PortFast**

S1(config)# interface fa0/1

S1(config-if)# switchport mode access

S1(config-if)# spanning-tree portfast

S1(config-if)# exit

S1(config)# spanning-tree portfast default

S1(config)# exit

**Configure BPDU Guard**

S1(config)# interface fa0/1

S1(config-if)# spanning-tree bpduguard enable

S1(config-if)# exit

S1(config)# spanning-tree portfast bpduguard default

S1(config)# end

S1# show spanning-tree summary

---

**Static Routing**

Router(config)# ip route network-address subnet-mask { ip-address | exit-intf [ip-address]} [distance]

Router(config)# ipv6 route ipv6-prefix/prefix-length {ipv6-address | exit-intf [ipv6-address]} [distance]

**IPv4 Fully Specified Static Route**

ip route 172.16.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2

**IPv6 Fully Specified Static Route**

R1(config)# ipv6 route 2001:db8:acad:1::/64 s0/1/0 fe80::2

Pozn.: Interface has to be specified for a link-local nexthop fe80::2

**IPv4 Directly Connected Static Route**

R1(config)# ip route 172.16.1.0 255.255.255.0 s0/1/0

**IPv6 Directly Connected Static Route**

R1(config)# ipv6 route 2001:db8:acad:1::/64 s0/1/0

**IPv4 Default Static Route**

Router(config)# ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf}

**IPv6 Default Static Route**

Router(config)# ipv6 route ::/0 {ipv6-address | exit-intf}

**Verify a Static Route**

show ip route static

show ip route network

show running-config | section ip route

**Floating Static Routes**

R1(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.2 **5**

R1(config)# ipv6 route ::/0 2001:db8:feed:10::2 **5**

**Administrative Distance**

| Route Source | Administrative Distance |
|---|---|
| Directly connected | 0 |
| Static route | 1 |
| EIGRP summary route | 5 |
| External BGP | 20 |
| Internal EIGRP | 90 |
| OSPF | 110 |
| IS-IS | 115 |
| RIP | 120 |
| External EIGRP | 170 |
| Internal BGP | 200 |

**Configure Static Host Routes**

Branch(config)# ip route 209.165.200.238 **255.255.255.255** 198.51.100.2

Branch(config)# ipv6 route 2001:db8:acad:2::238/**128** 2001:db8:acad:1::2

Branch(config)# exit

**SRWE PTSA Part 1 - ID 121**

---

**Addressing & VLAN Table (ID 121):**

Network: 172.31.0.0/16 | Router: R-1 | MLS: MULTI-1 | Switches: S-1, S-2, S-3

VLAN 10 (B1F1): 172.31.10.0/24 (GW 172.31.10.1)

VLAN 20 (B1F2): 172.31.20.0/24 (GW 172.31.20.1)

VLAN 30 (B1F4): 172.31.30.0/24 (GW 172.31.30.1)

VLAN 40 (B3): 172.31.40.0/24 (GW 172.31.40.1)

VLAN 50 (B4): 172.31.50.0/24 (GW 172.31.50.1)

VLAN 60 (B5): 172.31.60.0/24 (GW 172.31.60.1)

VLAN 99 (NetAdmin, native): 172.31.99.16/28 (GW 172.31.99.17, SVI S-3 172.31.99.18)

Links: R-1 <-> MULTI-1: 172.31.0.0/24 (.1 / .2) | R-1 <-> ISP: 209.165.201.0/30 (S0/1/0: .2)

**Default Gateways (IMPORTANT - Configure manually on PCs in Packet Tracer):**

- **Host 1 & Host 4** -> Default Gateway: **172.31.10.1** (VLAN 10)
- **Host 2 & Host 5** -> Default Gateway: **172.31.20.1** (VLAN 20)
- **Host 3 & Host 6** -> Default Gateway: **172.31.30.1** (VLAN 30)
- **Host 7** -> Default Gateway: **172.31.40.1** (VLAN 40)
- **Host 8** -> Default Gateway: **172.31.50.1** (VLAN 50)
- **Host 9** -> Default Gateway: **172.31.60.1** (VLAN 60)
- **Admin Host** -> Default Gateway: **172.31.99.17** (VLAN 99)

---

**Part 1: Basic Configuration (CLI: R-1)**

> en

# conf t

(config)# ip route 172.31.10.0 255.255.255.0 g0/0/0

(config)# ip route 172.31.20.0 255.255.255.0 g0/0/0

(config)# ip route 172.31.30.0 255.255.255.0 g0/0/0

(config)# no ip domain lookup

(config)# ena sec class

(config)# line con 0

(config-line)# pass cisco

(config-line)# login

(config-line)# exit

(config)# line vty 0 4

(config-line)# pass cisco

(config-line)# login

(config-line)# exit

(config)# banner motd #Authorized Access Only!#

(config)# hostname R-1

(config)# service password-encryption

(config)# int g0/0/0

(config-if)# desc Link to MULTI-1 G1/1/1

(config-if)# ip add 172.31.0.1 255.255.255.0

(config-if)# no shut

(config-if)# exit

(config)# int s0/1/0

(config-if)# desc Link to ISP

(config-if)# ip add 209.165.201.2 255.255.255.252

(config-if)# no shut

(config-if)# exit

**Part 5: Inter-VLAN Routing (CLI: R-1)**

(config)# int g0/0/1

(config-if)# desc Trunk to S-3 G0/1

(config-if)# no shut

(config-if)# exit

(config)# int g0/0/1.40

(config-subif)# desc Gateway for VLAN 40 (B3)

(config-subif)# encap dot 40

(config-subif)# ip add 172.31.40.1 255.255.255.0

(config-subif)# exit

(config)# int g0/0/1.50

(config-subif)# desc Gateway for VLAN 50 (B4)

(config-subif)# encap dot 50

(config-subif)# ip add 172.31.50.1 255.255.255.0

(config-subif)# exit

(config)# int g0/0/1.60

(config-subif)# desc Gateway for VLAN 60 (B5)

(config-subif)# encap dot 60

(config-subif)# ip add 172.31.60.1 255.255.255.0

(config-subif)# exit

(config)# int g0/0/1.99

(config-subif)# desc Gateway for VLAN 99 (NetAdmin, native)

(config-subif)# encap dot 99 native

(config-subif)# ip add 172.31.99.17 255.255.255.240

(config-subif)# exit

(config)# end

# copy run start

**Verification (CLI: R-1)**

# sh ip int br

# sh ip route

---

**Part 3: VLAN Configuration (CLI: MULTI-1)**

> en

# conf t

(config)# hostname MULTI-1

(config)# vlan 10

(config-vlan)# name B1F1

(config-vlan)# exit

(config)# vlan 20

(config-vlan)# name B1F2

(config-vlan)# exit

(config)# vlan 30

(config-vlan)# name B1F4

(config-vlan)# exit

**Part 4: EtherChannel & Trunking (CLI: MULTI-1)**

(config)# int range g1/0/1 - 2

(config-if-range)# channel-group 1 mode act

(config-if-range)# exit

(config)# int range g1/0/3 - 4

(config-if-range)# channel-group 2 mode act

(config-if-range)# exit

(config)# int po 1

(config-if)# sw mode trunk

(config-if)# sw nonegotiate

(config-if)# exit

(config)# int po 2

(config-if)# sw mode trunk

(config-if)# sw nonegotiate

(config-if)# exit

**Troubleshoot Po2:**

(config)# int range g1/0/3 - 4

(config-if-range)# no channel-group 2

(config-if-range)# sw mode trunk

(config-if-range)# sw nonegotiate

(config-if-range)# channel-group 2 mode act

(config-if-range)# shut

(config-if-range)# no shut

(config-if-range)# exit

**Part 5: Inter-VLAN Routing (CLI: MULTI-1)**

(config)# ip routing

(config)# int vlan 10

(config-if)# ip add 172.31.10.1 255.255.255.0

(config-if)# no shut

(config-if)# exit

(config)# int vlan 20

(config-if)# ip add 172.31.20.1 255.255.255.0

(config-if)# no shut

(config-if)# exit

(config)# int vlan 30

(config-if)# ip add 172.31.30.1 255.255.255.0

(config-if)# no shut

(config-if)# exit

(config)# int g1/1/1

(config-if)# no sw

(config-if)# ip add 172.31.0.2 255.255.255.0

(config-if)# no shut

(config-if)# exit

(config)# end

# copy run start

**Verification (CLI: MULTI-1)**

# sh eth sum

# sh int trunk

# sh ip route

Note: If Po2 member shows (s) or (I), enter 'shut' and 'no shut' on G1/0/3-4 of MULTI-1 and G0/1-2 of S-2.

---

**Part 3: VLAN Configuration & Access Ports (CLI: S-1)**

> en

# conf t

(config)# vlan 10

(config-vlan)# name B1F1

(config-vlan)# exit

(config)# vlan 20

(config-vlan)# name B1F2

(config-vlan)# exit

(config)# vlan 30

(config-vlan)# name B1F4

(config-vlan)# exit

(config)# int range f0/7 - 10

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 10

(config-if-range)# exit

(config)# int range f0/11 - 15

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 20

(config-if-range)# exit

(config)# int range f0/16 - 24

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 30

(config-if-range)# exit

**Part 4: EtherChannel & Trunking (CLI: S-1)**

(config)# int range g0/1 - 2

(config-if-range)# channel-group 1 mode act

(config-if-range)# exit

(config)# int range f0/5 - 6

(config-if-range)# channel-group 3 mode act

(config-if-range)# exit

(config)# int po 1

(config-if)# sw mode trunk

(config-if)# sw nonegotiate

(config-if)# exit

(config)# int po 3

(config-if)# sw mode trunk

(config-if)# sw nonegotiate

(config-if)# exit

(config)# end

# copy run start

**Verification (CLI: S-1)**

# sh eth sum

# sh int trunk

# sh vlan br

---

**Part 3: VLAN Configuration & Access Ports (CLI: S-2)**

> en

# conf t

(config)# vlan 10

(config-vlan)# name B1F1

(config-vlan)# exit

(config)# vlan 20

(config-vlan)# name B1F2

(config-vlan)# exit

(config)# vlan 30

(config-vlan)# name B1F4

(config-vlan)# exit

(config)# int range f0/7 - 10

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 10

(config-if-range)# exit

(config)# int range f0/11 - 15

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 20

(config-if-range)# exit

(config)# int range f0/16 - 24

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 30

(config-if-range)# exit

**Part 4: EtherChannel & Trunking (CLI: S-2)**

(config)# int range g0/1 - 2

(config-if-range)# channel-group 2 mode act

(config-if-range)# exit

(config)# int range f0/5 - 6

(config-if-range)# channel-group 3 mode act

(config-if-range)# exit

(config)# int po 2

(config-if)# sw mode trunk

(config-if)# sw nonegotiate

(config-if)# exit

(config)# int po 3

(config-if)# sw mode trunk

(config-if)# sw nonegotiate

(config-if)# exit

**Troubleshoot Po2 & duplex:**

(config)# int range g0/1 - 2

(config-if-range)# no channel-group 2

(config-if-range)# sw mode trunk

(config-if-range)# sw nonegotiate

(config-if-range)# dup auto

(config-if-range)# channel-group 2 mode act

(config-if-range)# shut

(config-if-range)# no shut

(config-if-range)# exit

(config)# end

# copy run start

**Verification (CLI: S-2)**

# sh eth sum

# sh int trunk

# sh vlan br

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**Part 2: Basic Configuration & SSH (CLI: S-3)**

> en

# conf t

(config)# int vlan 99

(config-if)# ip add 172.31.99.18 255.255.255.240

(config-if)# no shut

(config-if)# exit

(config)# ip default-gateway 172.31.99.17

(config)# ip domain-name acad.pt

(config)# ena sec class

(config)# username admin priv 15 sec C1sco123!

(config)# crypto key generate rsa general-keys modulus 1024

(config)# ip ssh ver 2

(config)# line vty 0 15

(config-line)# transport input ssh

(config-line)# login local

(config-line)# exit

**Part 3: VLAN Configuration & Access Ports (CLI: S-3)**

(config)# vlan 40

(config-vlan)# name B3

(config-vlan)# exit

(config)# vlan 50

(config-vlan)# name B4

(config-vlan)# exit

(config)# vlan 60

(config-vlan)# name B5

(config-vlan)# exit

(config)# vlan 99

(config-vlan)# name NetAdmin

(config-vlan)# exit

(config)# int range f0/1 - 5

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 40

(config-if-range)# exit

(config)# int range f0/6 - 10

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 50

(config-if-range)# exit

(config)# int range f0/11 - 15

(config-if-range)# sw mode acc

(config-if-range)# sw acc vlan 60

(config-if-range)# exit

(config)# int f0/24

(config-if)# sw mode acc

(config-if)# sw acc vlan 99

(config-if)# exit

**Part 4: Trunking Configuration (CLI: S-3)**

(config)# int g0/1

(config-if)# sw mode trunk

(config-if)# sw trunk native vlan 99

(config-if)# sw nonegotiate

(config-if)# exit

(config)# end

# copy run start

**Verification (CLI: S-3)**

# sh ip int br

# sh int trunk

# ping 172.31.99.17

Verify SSH from Admin Host: ssh -l admin 172.31.99.18 (pass C1sco123!)

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